BAHIR DAR INSTITUTE OF TECHNOLOGY

Faculty of Computing

Department of Software Engineering

Operating System and System Programing (OSSP) Individual Assignment

Installation of BlackArch Linux OS

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1. Introduction

BlackArch Linux is an open-source penetration testing distribution based on Arch Linux, specifically designed for security researchers and penetration testers. It serves as a comprehensive toolkit, offering a repository of over 2,800 specialized tools for various cyber security tasks, including web application testing, digital forensics, and network analysis. The distribution follows a rolling release model, ensuring that users always have access to the latest software updates and security patches. The "Slim ISO" version is a streamlined edition that provides a functional desktop environment with essential tools, making it an ideal choice for users who require a faster installation process while maintaining the flexibility to add more tools as needed.

The motivation behind using BlackArch Linux lies in its lightweight nature and the efficiency of the Arch Linux ecosystem. Unlike other distributions that may include unnecessary bloatware, BlackArch allows for a highly customized environment. By utilizing Oracle VM VirtualBox, users can create an isolated laboratory environment, enabling them to conduct security experiments and learn penetration testing techniques without risking the stability or security of their primary host operating system.

2. Objectives

The primary objective of this documentation is to provide a clear, step-by-step guide for the successful installation of BlackArch Linux within a virtualized environment. This process aims to achieve the following:

- **Installation**: Install BlackArch operating system in a virtual machine.
- **System configuration**: Optimize virtual machine settings for the best performance of an Arch-based guest system.
- **User Management**: Implement proper user account setup as per administrative requirements.
- **Educational Foundation**: Enhance understanding of Linux installation processes, partitioning, and file system management.
- **Troubleshooting**: Identify and solve installation problems.

3. Requirements

The successful installation of BlackArch Linux requires a combination of hardware and software resources.

3.1. Hardware Requirements

To successfully install BlackArch Linux in a virtual environment, the following hardware specifications are required:

- **Processor (CPU)**: A 64-bit processor with virtualization support.
- **Memory (RAM)**: Minimum of 2 GB. 4GB or higher recommended for smoother performance during installation and operation.
- **Storage (Hard Disk)**: At least 40 GB of free disk space to allocate to the virtual machine.
- **Display**: Standard monitor capable of displaying graphical user interface.
- **Internet Connection**: Required during installation for downloading and synchronizing system packages.

3.2. Software Requirements

- **Virtualization Software**: Oracle VM VirtualBox used to create and manage the virtual machine environment.
- **Operating System**: BlackArch Linux Slim ISO used as installation source for the virtual machine.
- **Host Operating System**: Any modern OS such as Windows, Linux, or MacOS capable of running VirtualBox.

4. Installation Steps

Step 1: Install Virtualization Software

- Downloaded and installed Oracle VM VirtualBox

Step 2: Create and Configure Virtual Machine

- Opened VirtualBox and selected “New”. Enter "BlackArch" as the name, select "Linux" as the type, and "Arch Linux (64-bit)" as the version.
- Assigned at least 4096 MB of RAM and 2 CPU cores to ensure the system operates smoothly during the installation.

- Created a Virtual Disk Image (VDI) with at least 40 GB of space, using “Dynamically allocated” storage.

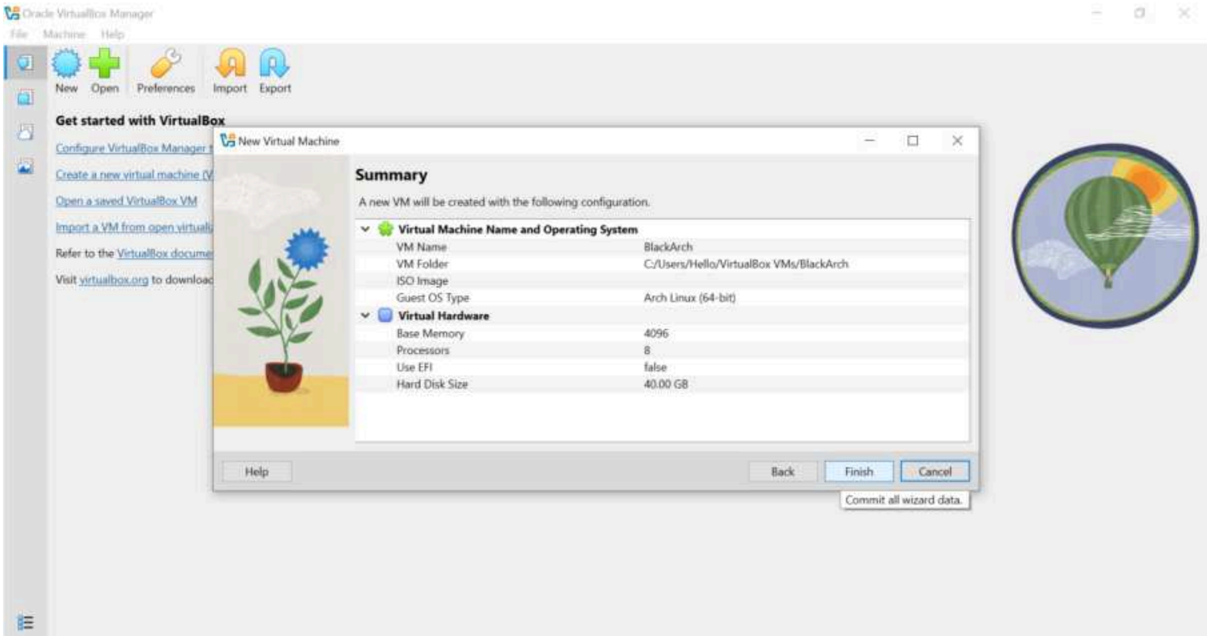

Figure 1: Creating and configuring BlackArch virtual machine

Step 3: Attach Installation ISO

- Navigated to Settings > Storage. Selected the empty optical drive and browsed for the downloaded blackarch-slim-64-bit.iso file.

Step 4: Configure Network Settings

- Ensures the network adapter is enabled and set to “NAT”.

Step 5: Boot the System

- Started the VM and selected “BlackArch Linux Slim (x86_64, BIOS)” from the boot menu.

- Logged in using default credentials
    - **Username**: liveuser
    - **Password**: blackarch

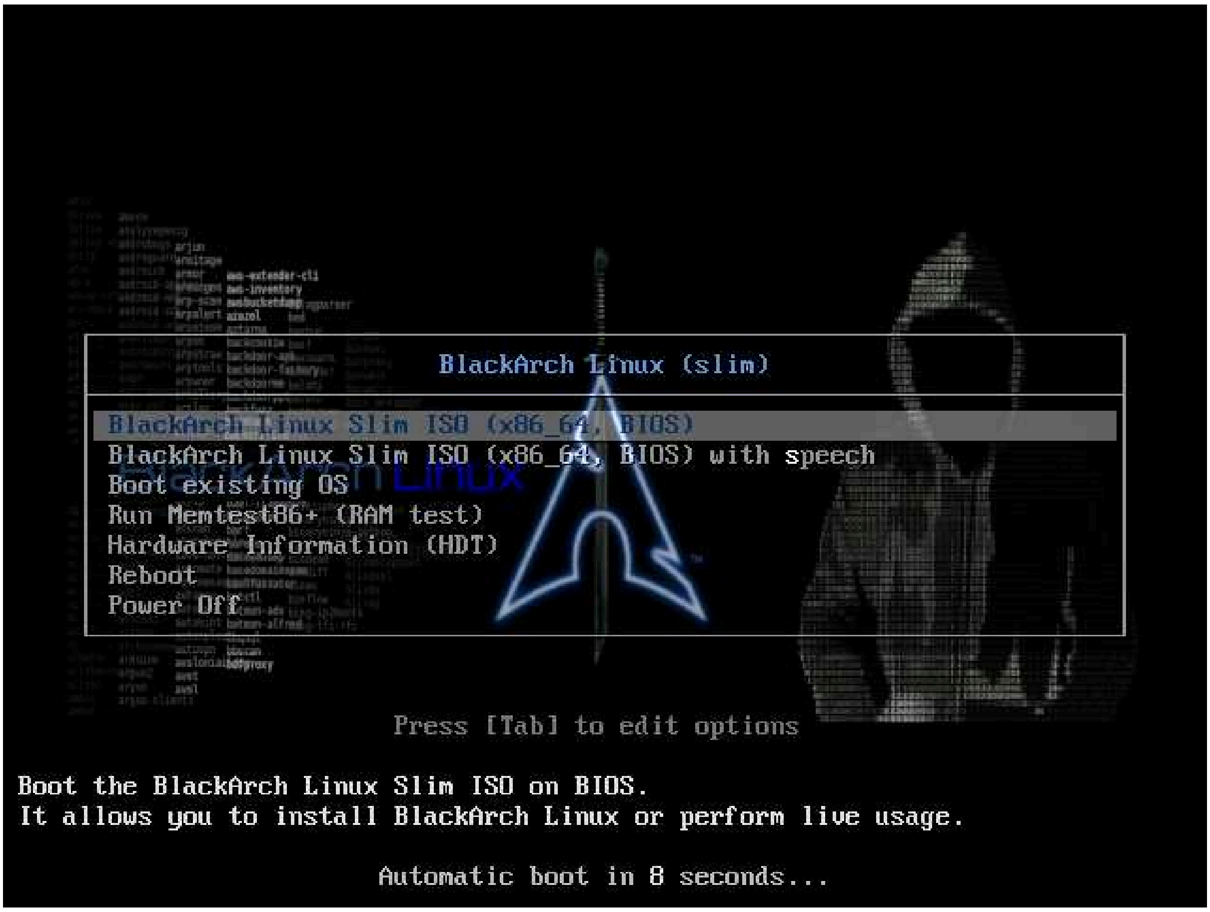

Figure 2: Booting BlackArch

Figure 3: Logging in using default credential

Step 6: Launch Installer

- Accessed graphical desktop environment and opened “Install BlackArch”.

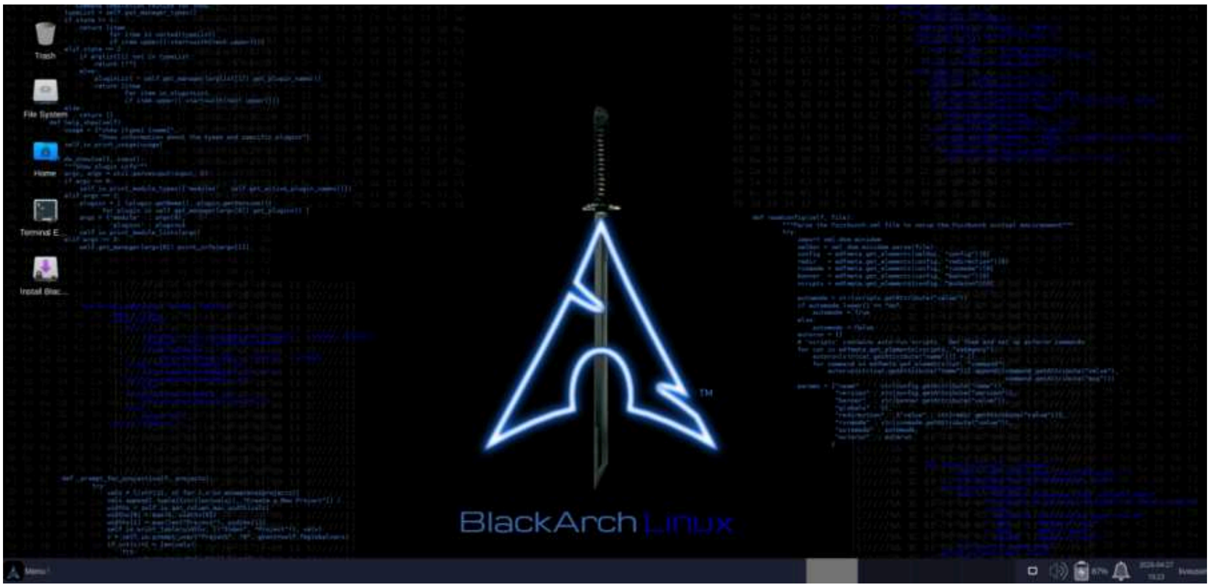

Figure 4: BlackArch desktop with installer icon

Step 7: Configure Installation Settings and Disk Partition

- Selected language, region, and keyboard layout as provided by the BlackArch Linux installer.
- Selected “Erase Disk” (automatic partitioning) and used default filesystem (ext4).

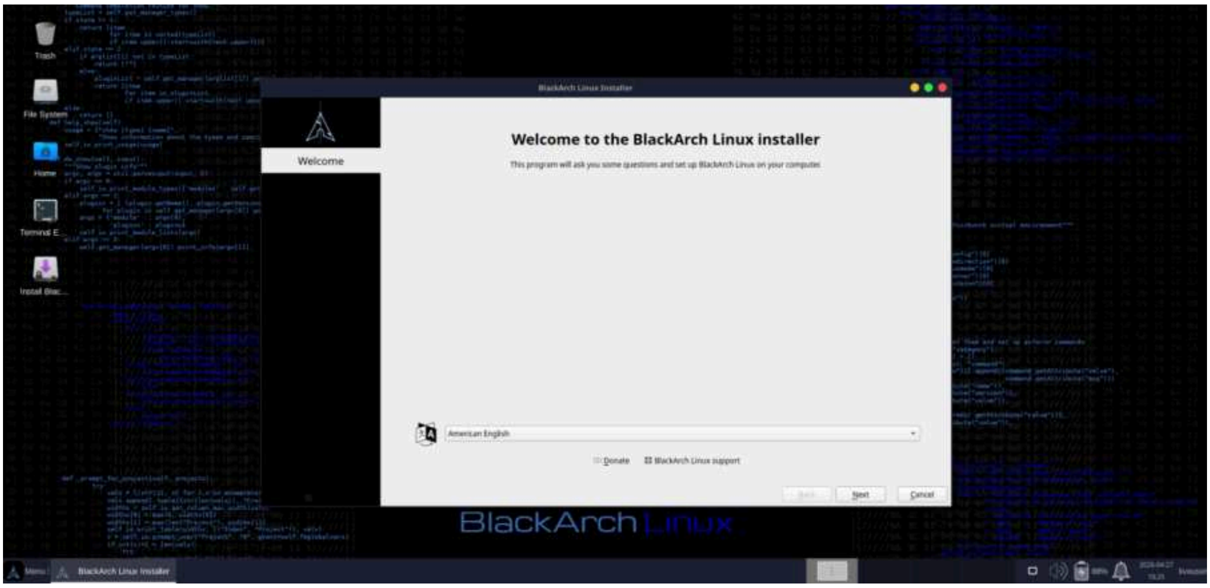

Figure 5: Selecting language

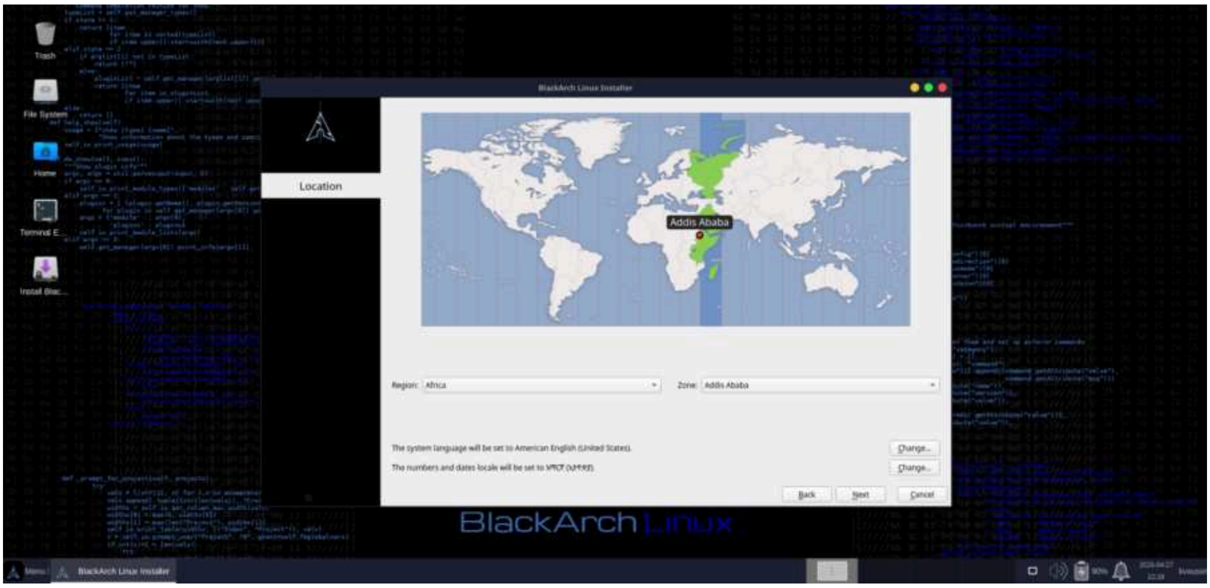

Figure 6: Selecting region and zone

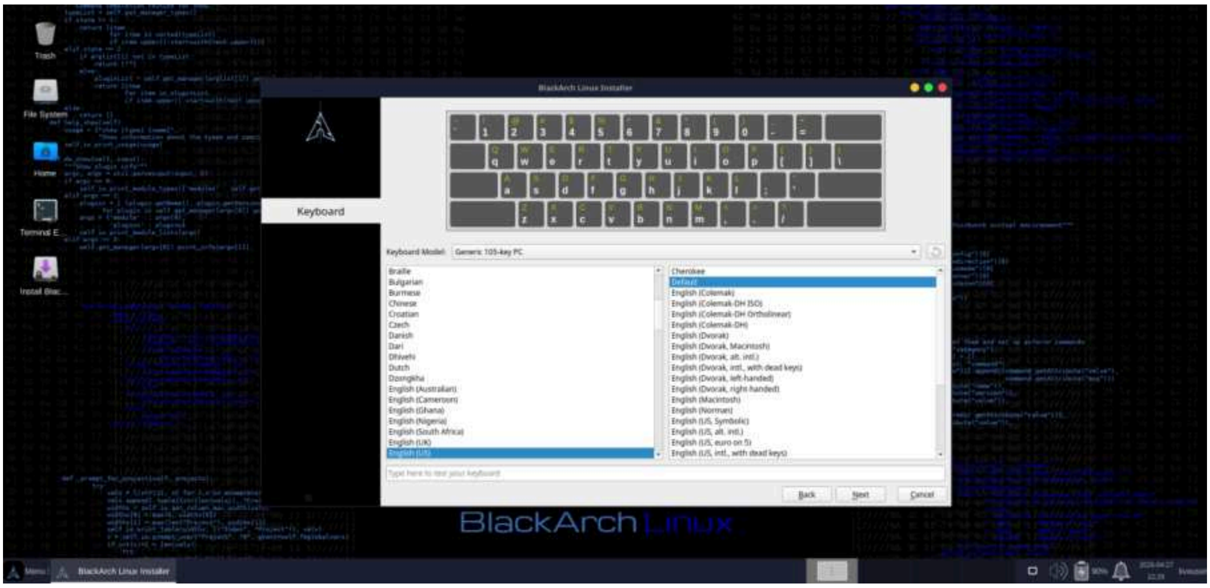

Figure 7: Selecting keyboard layout

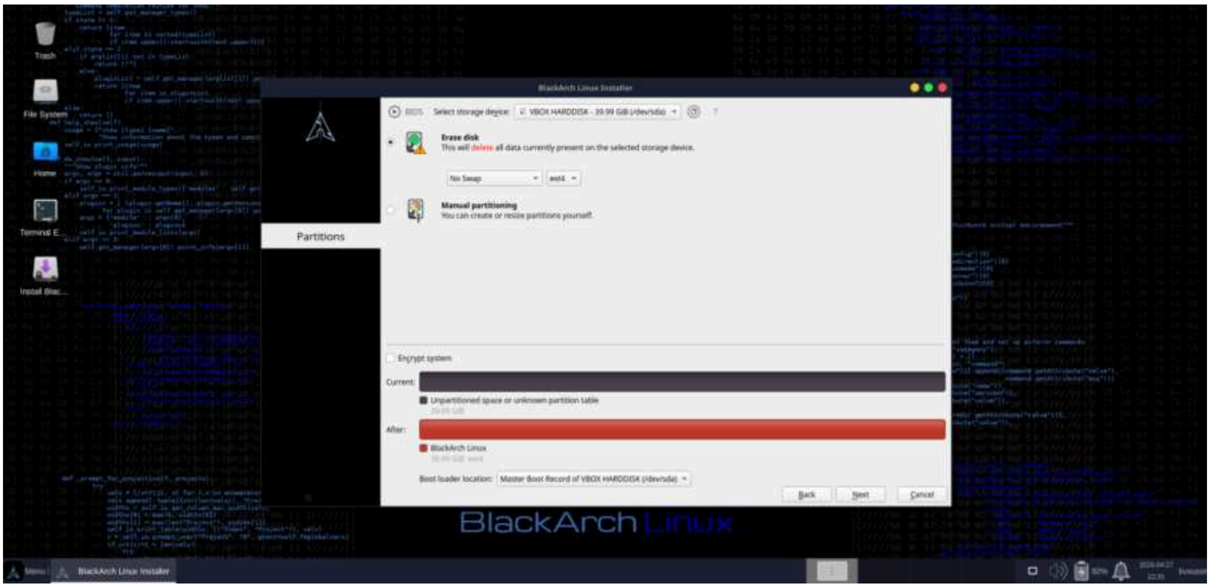

Figure 8: Selecting erase disk and ext4

Step 8: User Account Creation

- Entered full name as required and created username and password.

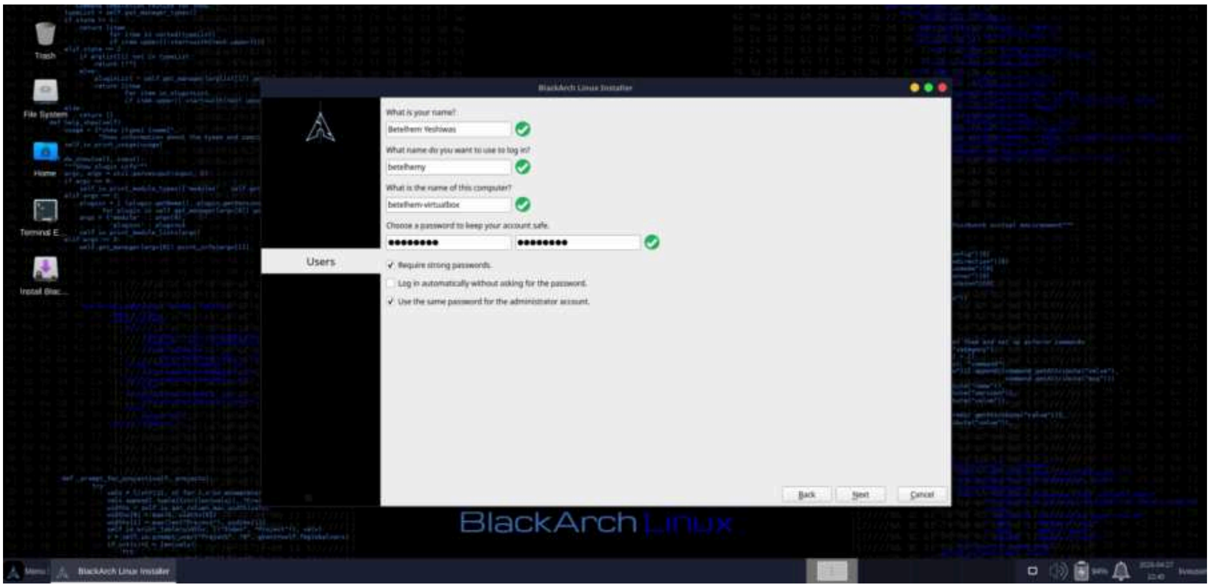

Figure 9: Creating an account using my full name

Step 9: Installation Process

- Reviewed summary settings and started installation.

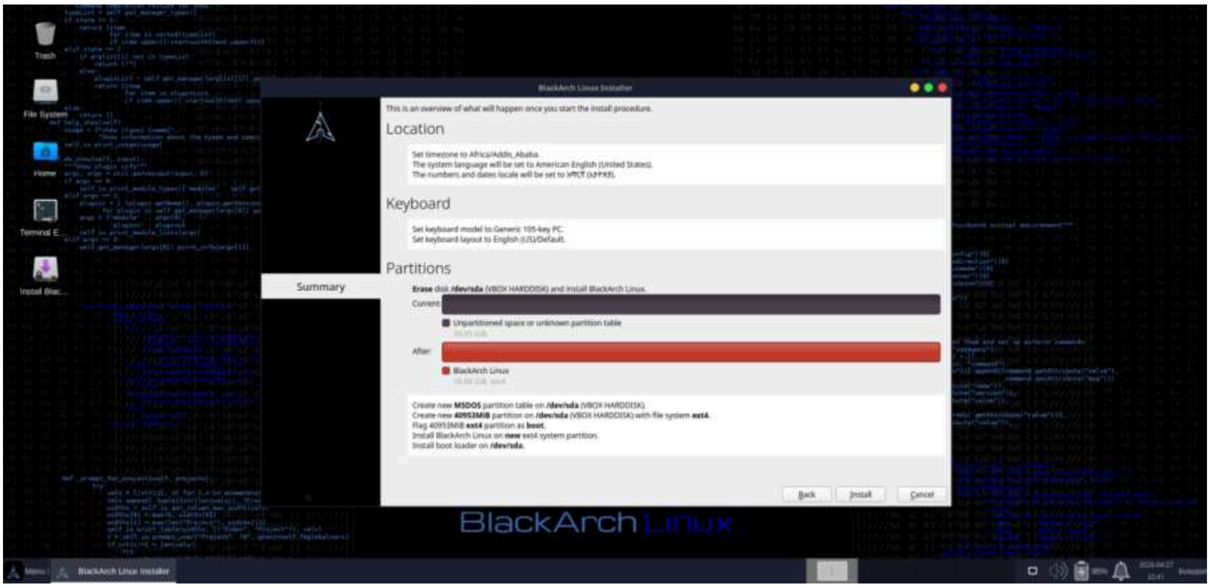

Figure 10: Installation summary

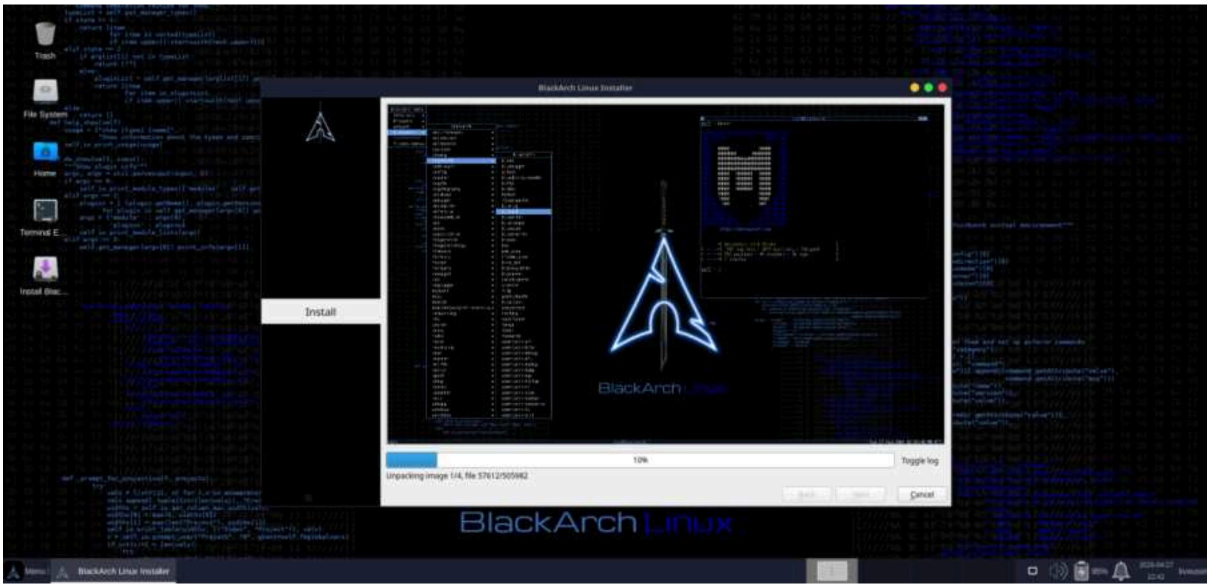

Figure 11: Installation on progress

Step 10: Reboot and Final Setup

- Rebooted VM after installation is done and removed ISO from virtual device then logged into the installed system using the name and password provided during the installation process.

Figure 12: Logging into the installed BlackArch system

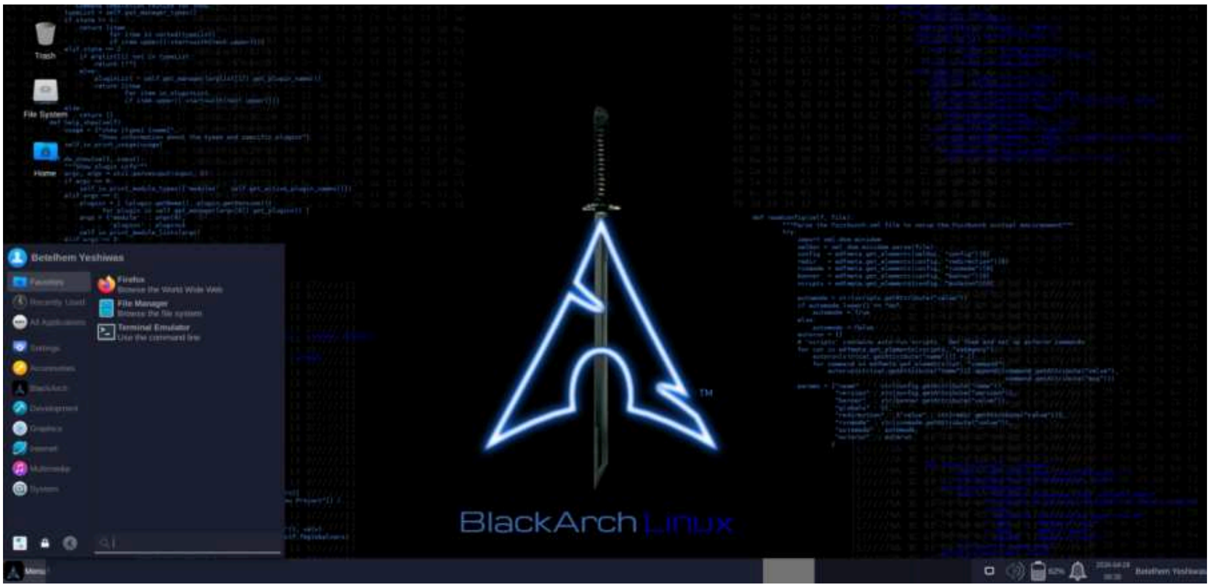

Figure 13: BlackArch desktop after installation

5. Issues (Problems Faced)

During the installation of BlackArch Linux in Oracle VM VirtualBox, a critical error occurred at the package installation stage.

Error Message: “The package manager could not prepare updates. The command pacman returned error code 1.”

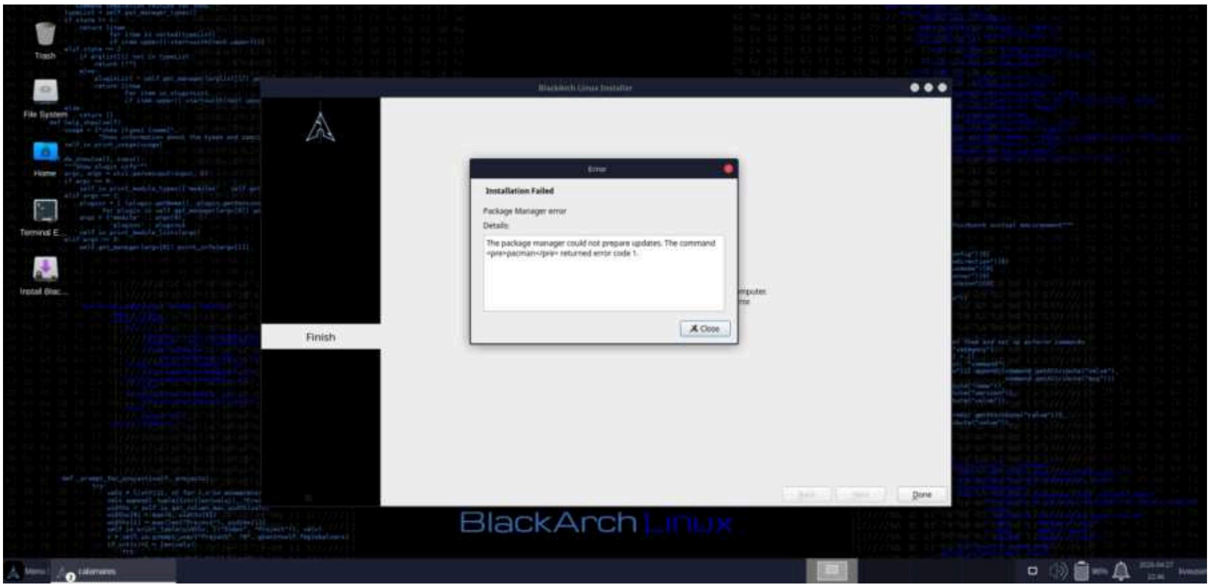

Figure 14: Installation error

The *pacman* error code 1 was a critical installation issue caused by failure in package synchronization from online repositories. Despite having a working internet connection, the installer could not reliably communicate with mirror servers, leading to installation failure. In Arch based systems like BlackArch, *pacman* is the default package manager. It is responsible for installing system packages, synchronizing package databases and downloading required files during OS installation. The error code 1 indicates that *pacman* was unable to complete its operation successfully.

6. Solution

To resolve the error encountered during the installation, several troubleshooting methods were performed.

Verifying basic system functionality by confirming that the virtual machine had internet access by using “ping –c 3 google.com” and restarted the system to clear temporary installation issues. But these methods didn’t solve the problem so an alternative troubleshooting method was explored using Linux commands.

- “sudo su” was used to gain root access.

- “ls” was used to navigate system directories.

- “nano /etc/pacman.conf” was used to inspect package manager settings

- “nano /etc/calamares/modules/packages.conf” was used to modify behavior by setting “update_db: false” to prevent installer from updating package database, avoiding *pacman* errors.

- “sudo pacman –Syy” was used to force refresh the package database to resolve synchronization issues during installation.

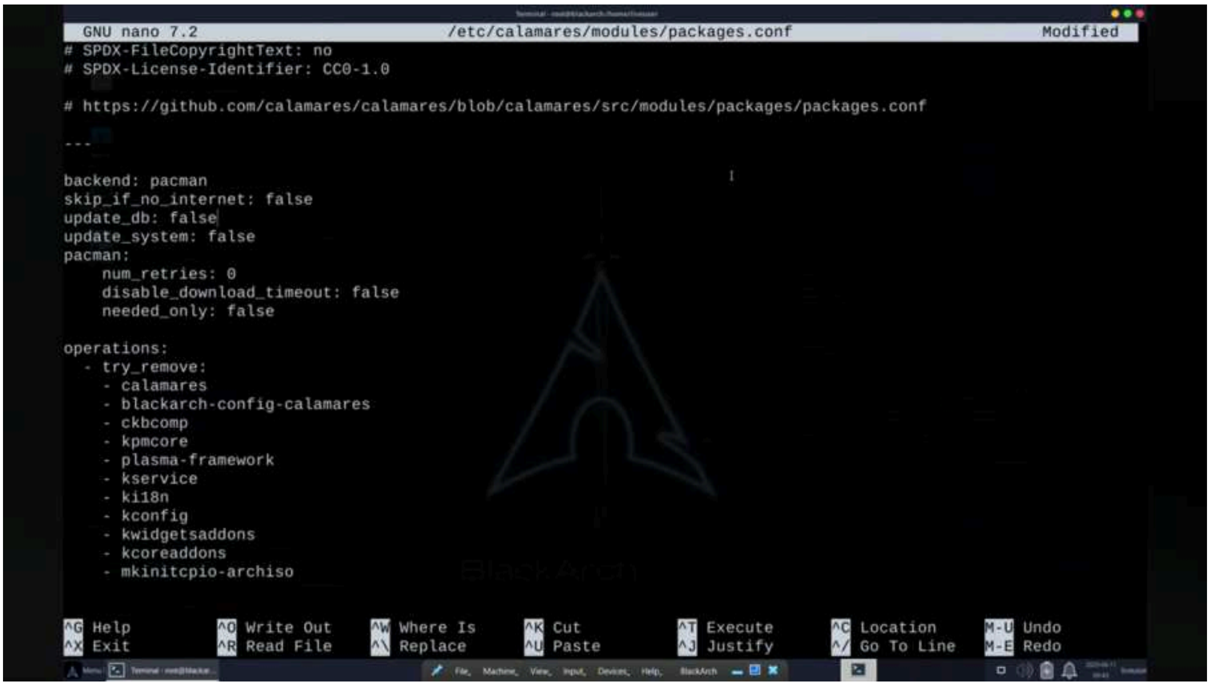

Figure 15: setting update_db: false

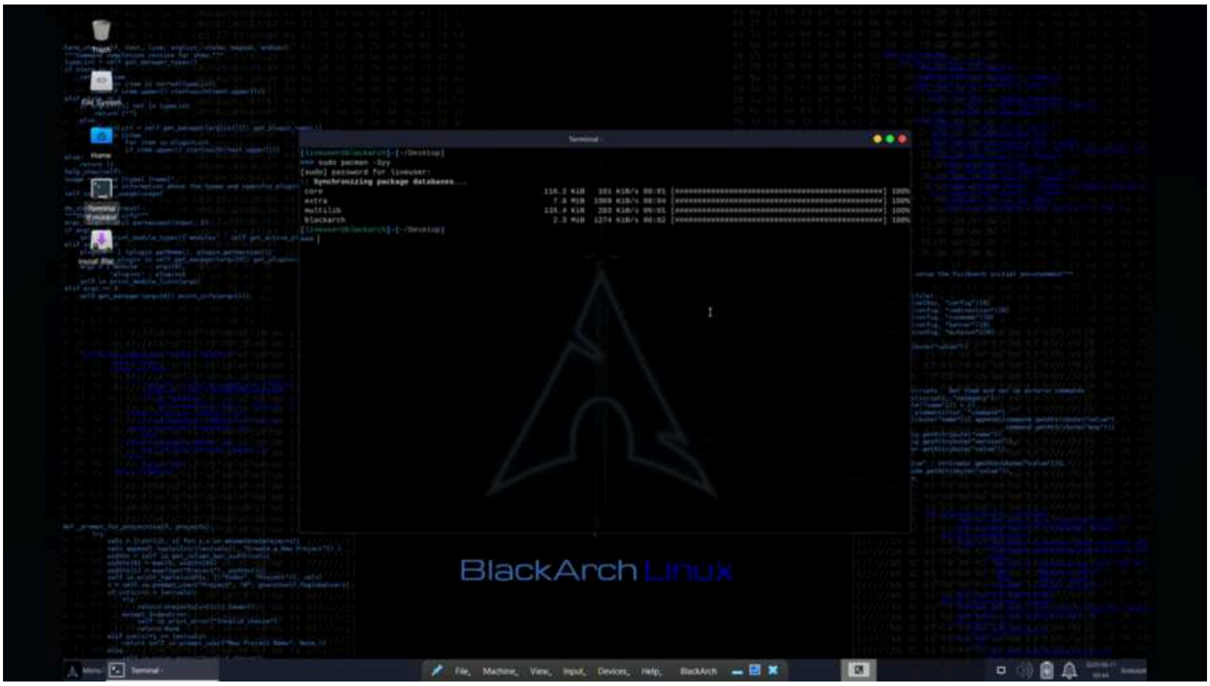

Figure 16: synchronizing package database

This method solved the error and BlackArch was successfully installed.

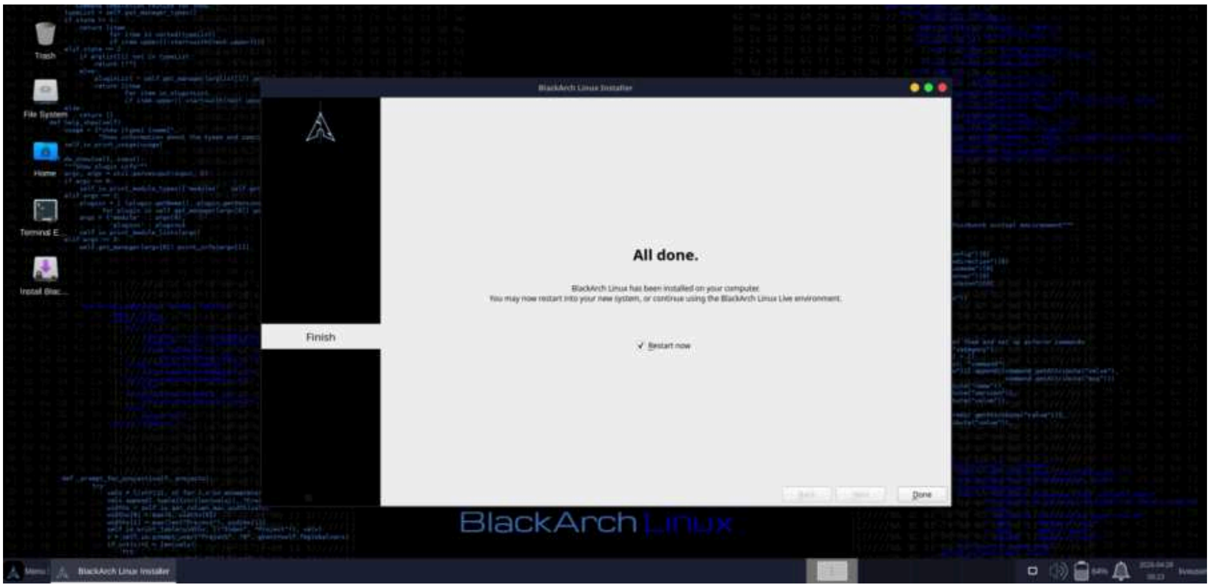

Figure 17: Successful installation of BlackArch

7. Filesystem Support

Choosing the correct filesystem is critical for system performance and data integrity. BlackArch supports various filesystems, including **ext4, Btrfs, ZFS,** and **XFS**.

For this installation, **ext4** is utilized. It is the default filesystem for most Linux distributions, highly stable and reliable, provides good performance, supports large files and partitions, and fully compatible with BlackArch. **ext4** is widely recommended for beginners and virtual machine environments due to its simplicity and low risk of errors.

Btrfs offers advanced features like snapshots (useful for reverting system changes), it carries a higher risk of fragmentation in a virtualized environment. It is complex to configure and not ideal for beginners or simple installations.

ZFS provides superior data integrity but requires significant RAM overhead, which may not be ideal for a standard virtual machine setup.

Xfs is a high performance filesystem designed for handling large amount of data and fast read/write performance. It is less flexible for resizing and not common for beginner setup.

The **ext4** filesystem was chosen for this installation due to its reliability, compatibility, and ease of use. While other file systems offer advanced features, ext4 remains the most suitable option for standard installations and virtual machine environments.

8. Advantages and Disadvantages

The use of BlackArch Linux on VirtualBox presents several trade-offs that users should consider.

Advantages

- **Vast Tool Repository**: Access to over 2,800 tools without manual installation.
- **Isolation**: Testing security tools in a VM prevents accidental damage to the host system.
- **Rolling Release**: The system is always up-to-date with the latest security research.
- **Efficiency**: Arch-based systems are generally faster and consume fewer resources than Debian-based alternatives.
- **Education**: useful for cybersecurity learning.

Disadvantages

- **Complexity**: Arch Linux has a steeper learning curve compared to Ubuntu or Kali.
- **Stability Risks**: Rolling updates can occasionally break system configurations or tool dependencies.
- **Storage Requirements**: A full installation can consume significant disk space (up to 50 GB+).

9. Conclusion

The installation of BlackArch Linux on Oracle VM VirtualBox provides a powerful and flexible platform for cybersecurity professionals. By following the structured installation process and selecting the reliable ext4 filesystem, BlackArch was successfully installed. The process provided practical experience in virtualization, system installation, and troubleshooting.

10. Future Outlook and Recommendations

As cybersecurity threats evolve, the demand for specialized distributions like BlackArch will continue to grow. For future improvements, it is recommended to:

- **Implement Snapshots**: Regularly take VirtualBox snapshots before performing major system updates.
- **Use Shared Folders**: Set up VirtualBox Guest Additions to enable easy file transfer between the host and the VM.

11. Virtualization in Modern Operating Systems

Virtualization is the process of creating a software-based (virtual) representation of something, such as virtual applications, servers, storage, and networks. It is the single most effective way to reduce IT expenses while boosting efficiency and agility for all size businesses.

The "What"

At its core, virtualization uses software to simulate hardware functionality and create a virtual computer system. This allows IT organizations to run more than one virtual system—and multiple operating systems and applications—on a single server.

The "Why"

- **Resource Optimization**: Most physical servers operate at less than 15% capacity; virtualization allows multiple workloads to share a single physical resource.
- **Security and Isolation**: If one virtual machine is compromised, the others remain protected, making it ideal for malware analysis and penetration testing.
- **Disaster Recovery**: Virtual machines can be easily backed up, moved, and restored, significantly reducing downtime.

The "How"

Virtualization relies on a piece of software called a **Hypervisor**. The hypervisor sits between the physical hardware and the virtual machines, dynamically allocating resources (CPU, RAM, Disk) to each "Guest" OS. There are two types: **Type 1 (Bare Metal)**, which runs directly on the hardware (e.g., VMware ESXi), and **Type 2 (Hosted)**, which runs as an application on a host OS (e.g., Oracle VM VirtualBox).